

Lecture 16

Labor Supply Choice

ECO303

Allocation of Time

- Individuals must decide how to allocate the fixed amount of time they have
- We will initially assume that there are only two uses of an individual's time
 - engaging in market work at a real wage rate of w
 - leisure (nonwork)

Allocation of Time

- Assume that an individual's utility depends on consumption (c) and hours of leisure (h)

$$\text{utility} = U(c, h)$$

- In seeking to maximize utility, the individual is bound by two constraints

$$l + h = 24$$

$$c = wl$$

Allocation of Time

- Combining the two constraints, we get

$$c = w(24 - h)$$

$$c + wh = 24w$$

- An individual has a “full income” of $24w$
 - may spend the full income either by working (for real income and consumption) or by not working (enjoying leisure)
- The opportunity cost of leisure is w

Utility Maximization

- The individual's problem is to maximize utility subject to the full income constraint
- Setting up the Lagrangian

$$\mathbf{L} = U(c,h) + \lambda(24w - c - wh)$$

- The first-order conditions are

$$\partial \mathbf{L} / \partial c = \partial U / \partial c - \lambda = 0$$

$$\partial \mathbf{L} / \partial h = \partial U / \partial h - \lambda = 0$$

Utility Maximization

- Dividing the two, we get

$$\frac{\partial U / \partial h}{\partial U / \partial c} = w = MRS(h \text{ for } c)$$

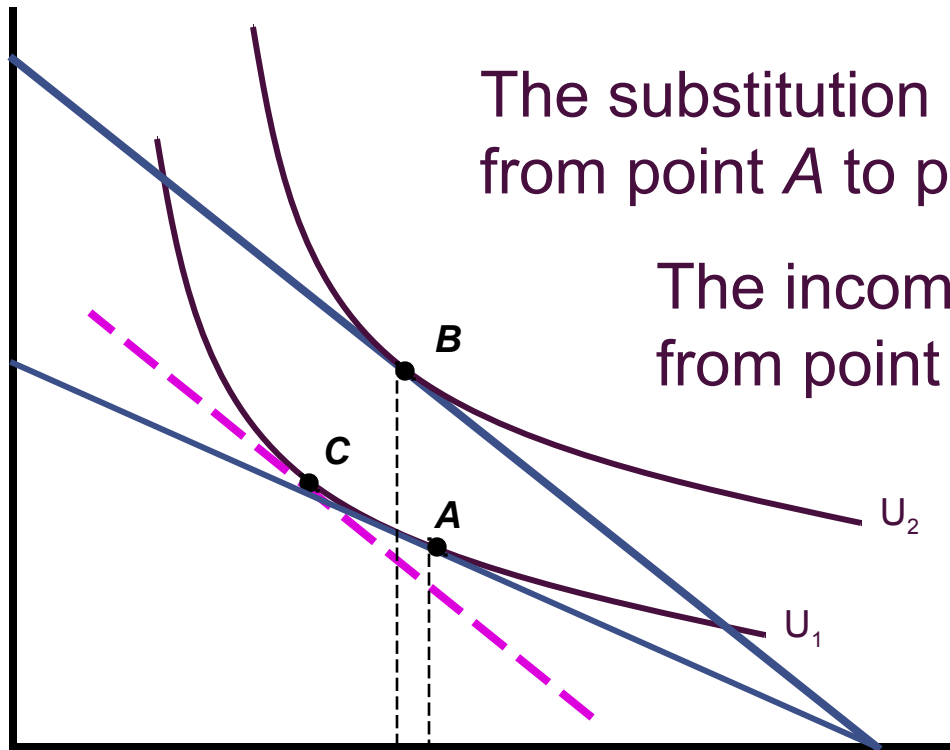
- To maximize utility, the individual should choose to work that number of hours for which the *MRS* (of *h* for *c*) is equal to *w*
 - to be a true maximum, the *MRS* (of *h* for *c*) must be diminishing

Income and Substitution Effects

- Both a substitution effect and an income effect occur when w changes
 - when w rises, the price of leisure becomes higher and the individual will choose less leisure
 - because leisure is a normal good, an increase in w leads to an increase in leisure
- The income and substitution effects move in opposite directions

Income and Substitution Effects

Consumption



The substitution effect is the movement from point *A* to point *C*

The income effect is the movement from point *C* to point *B*

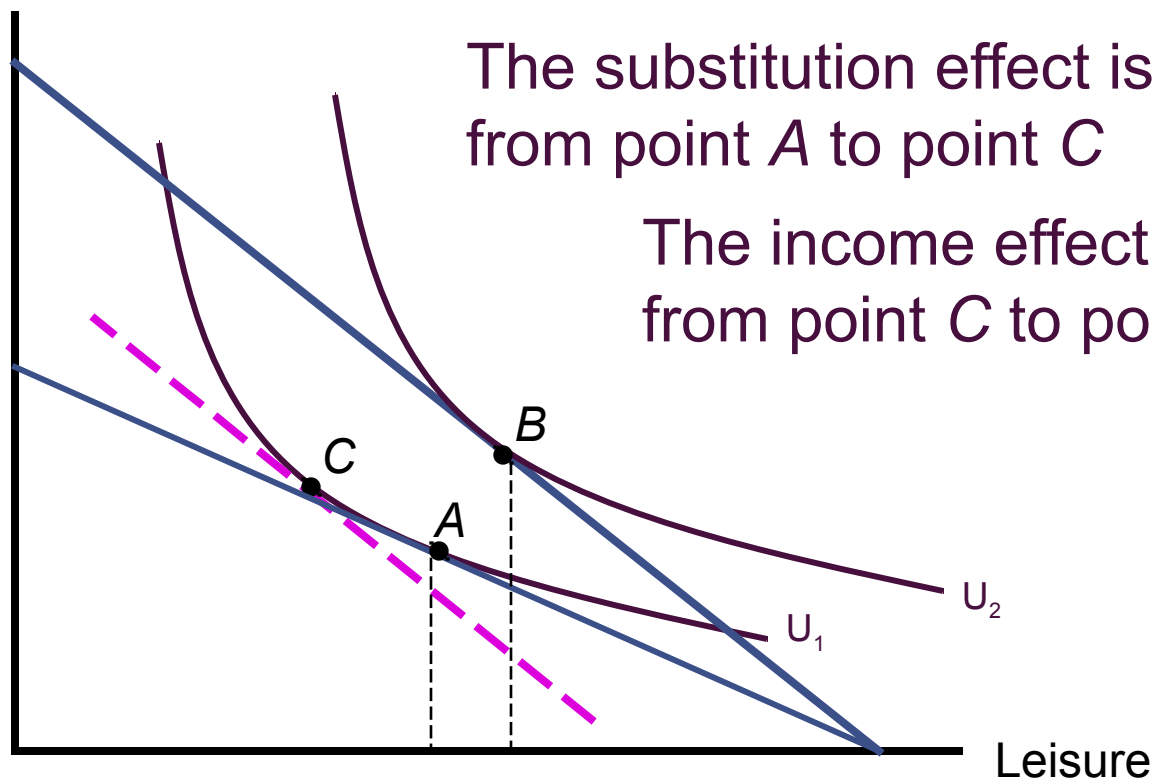
The individual chooses less leisure as a result of the increase in w

Leisure

substitution effect > income effect

Income and Substitution Effects

Consumption



substitution effect < income effect

A Mathematical Analysis of Labor Supply

- We will start by amending the budget constraint to allow for the possibility of nonlabor income

$$c = wl + n$$

- Maximization of utility subject to this constraint yields identical results
 - as long as n is unaffected by the labor-leisure choice

A Mathematical Analysis of Labor Supply

- The only effect of introducing nonlabor income is that the budget constraint shifts out (or in) in a parallel fashion
- We can now write the individual's labor supply function as $l(w,n)$
 - hours worked will depend on both the wage and the amount of nonlabor income
 - since leisure is a normal good, $\partial l / \partial n < 0$

Dual Statement of the Problem

- The dual problem can be phrased as choosing levels of c and h so that the amount of expenditure ($E = c - wl - n$) required to obtain a given utility level (U_0) is as small as possible
 - solving this minimization problem will yield exactly the same solution as the utility maximization problem

Dual Statement of the Problem

- A small change in w will change the minimum expenditures required by

$$\partial E / \partial w = -l$$

- this is the extent to which labor earnings are increased by the wage change

Dual Statement of the Problem

- This means that a labor supply function can be calculated by partially differentiating the expenditure function
 - because utility is held constant, this function should be interpreted as a “compensated” (constant utility) labor supply function

$$l^c(w, U)$$

Slutsky Equation of Labor Supply

- The expenditures being minimized in the dual expenditure-minimization problem play the role of nonlabor income in the primary utility-maximization problem

$$l^c(w, U) = l[w, E(w, U)] = l(w, n)$$

- Partial differentiation of both sides with respect to w gives us

$$\frac{\partial l^c}{\partial w} = \frac{\partial l}{\partial w} + \frac{\partial l}{\partial E} \cdot \frac{\partial E}{\partial w}$$

Slutsky Equation of Labor Supply

- Substituting for $\partial E / \partial w$, we get

$$\frac{\partial l^c}{\partial w} = \frac{\partial l}{\partial w} - l \frac{\partial l}{\partial E} = \frac{\partial l}{\partial w} - l \frac{\partial l}{\partial n}$$

- Introducing a different notation for l^c , and rearranging terms gives us the Slutsky equation for labor supply:

$$\frac{\partial l}{\partial w} = \frac{\partial l}{\partial w} \Big|_{U=U_0} + l \frac{\partial l}{\partial n}$$

Cobb-Douglas Labor Supply

- Suppose that utility is of the form

$$U = c^\alpha h^\beta, \text{ where } \alpha + \beta = 1$$

- The budget constraint is

$$c = wl + n$$

and the time constraint is

$$l + h = 1$$

- note that we have set maximum work time to 1 hour for convenience

Cobb-Douglas Labor Supply

- The Lagrangian expression for utility maximization is

$$\mathbf{L} = c^\alpha h^\beta + \lambda(w + n - wh - c)$$

- First-order conditions are

$$\partial \mathbf{L} / \partial c = \alpha c^{-\beta} h^\beta - \lambda = 0$$

$$\partial \mathbf{L} / \partial h = \beta c^\alpha h^{-\alpha} - \lambda w = 0$$

$$\partial \mathbf{L} / \partial \lambda = w + n - wh - c = 0$$

Cobb-Douglas Labor Supply

- Dividing the first by the second yields

$$\frac{\alpha h}{\beta c} = \frac{\partial h}{(1-\alpha)c} = \frac{1}{w}$$

$$wh = \frac{1-\alpha}{\alpha} \cdot c$$

Cobb-Douglas Labor Supply

- Substitution into the full income constraint yields

$$c = \alpha(w + n)$$

$$h = \beta(w + n)/w$$

- the person spends α of his income on consumption and $\beta = 1 - \alpha$ on leisure
- the labor supply function is

$$l(w, n) = 1 - h = (1 - \beta) - \frac{\beta n}{w}$$

Cobb-Douglas Labor Supply

- Note that if $n = 0$, the person will work $(1-\beta)$ of each hour no matter what the wage is
 - the substitution and income effects of a change in w offset each other and leave l unaffected

Cobb-Douglas Labor Supply

- If $n > 0$, $\partial l / \partial w > 0$
 - the individual will always choose to spend βn on leisure
 - Since leisure costs w per hour, an increase in w means that less leisure can be bought with n

Cobb-Douglas Labor Supply

- Note that $\partial l / \partial n < 0$
 - an increase in nonlabor income allows this person to buy more leisure
 - income transfer programs are likely to reduce labor supply
 - lump-sum taxes will increase labor supply

CES Labor Supply

- Suppose that the utility function is

$$U(c, h) = \frac{c^\delta}{\delta} + \frac{h^\delta}{\delta}$$

- Budget share equations are given by

$$s_c = \frac{c}{w+n} = \frac{1}{(1+w^\kappa)}$$

$$s_h = \frac{wh}{w+n} = \frac{1}{(1+w^{-\kappa})}$$

– where $\kappa = \delta/(\delta-1)$

CES Labor Supply

- Solving for leisure gives

$$h = \frac{w + n}{w + w^{1-\kappa}}$$

and

$$l(w, n) = 1 - h = \frac{w^{1-\kappa} - n}{w + w^{1-\kappa}}$$